

PSEUDOCODE WRITING

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QUESTIONS

Q1

Student Management System
[BL3 – Apply]
10 marks

Q2

Library Book Management [BL3 – Apply]
10 marks

Q3

Employee Salary Calculation
[BL3 – Apply]
10 marks

Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6

Q6 6. Bank Account Hierarchy [BL4 – Analyze]

10 marks

Develop pseudocode for a BankAccount superclass and two subclasses — SavingsAccount (with interest calculation) and CurrentAccount (with overdraft facility). Analyze and implement:

- Inheritance of common attributes (accNo, balance, holder)
- Method overriding for withdrawal behavior
- How overdraft protection differs from savings limit

Compare the behavior of withdraw() across both subclasses and justify the design choices.

OOP Concepts: Inheritance | Method Overriding | Polymorphism

Your Answer

CLASS BankAccount

ATTRIBUTES:

accNo : STRING

balance : REAL

holder : STRING

CONSTRUCTOR (accNo, balance, holder)

this.accNo = accNo

this.balance = balance

this.holder = holder

[BL3 – Apply]
10 marks

Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6

6. Bank Account Hierarchy [BL4
– Analyze]
10 marks

Q7

7. Hospital Patient Record
System [BL4 – Analyze]
10 marks

Q8

8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

Q9

9. University Staff Hierarchy
[BL4 – Analyze]
10 marks

Your Answer

```
METHOD deposit(amount : REAL)
    IF amount > 0 THEN
        balance = balance + amount
    ENDIF

METHOD withdraw(amount : REAL)
    IF amount > 0 THEN
        IF balance >= amount THEN
            balance = balance - amount
            PRINT "Withdrawal successful. New balance: ", balance
        ELSE
            PRINT "Insufficient balance"
        ENDIF
    ENDIF
ENDCLASS

CLASS SavingsAccount EXTENDS BankAccount
    ATTRIBUTES:
        interestRate : REAL

    CONSTRUCTOR (accNo, balance, holder, interestRate)
        CALL BankAccount.CONSTRUCTOR(accNo, balance, holder)
        this.interestRate = interestRate
```

Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks



Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks



Q6

6. Bank Account Hierarchy [BL4
– Analyze]
10 marks



Q7

7. Hospital Patient Record
System [BL4 – Analyze]
10 marks



Q8

8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks



Q9

9. University Staff Hierarchy
[BL4 – Analyze]
10 marks



Q10



```
this.interestRate = interestRate
```

```
METHOD calculateInterest() : REAL
```

```
RETURN balance * interestRate
```

```
METHOD withdraw(amount : REAL)
```

```
IF amount > 0 THEN
```

```
IF balance >= amount THEN
```

```
balance = balance - amount
```

```
PRINT "Savings withdrawal successful. New balance: ", balance
```

```
ELSE
```

```
PRINT "Savings: Insufficient balance (overdraft not allowed)"
```

```
ENDIF
```

```
ENDIF
```

```
ENDCLASS
```

```
CLASS CurrentAccount EXTENDS BankAccount
```

```
ATTRIBUTES:
```

```
overdraftLimit : REAL
```

```
CONSTRUCTOR (accNo, balance, holder, overdraftLimit)
```

```
CALL BankAccount.CONSTRUCTOR(accNo, balance, holder)
```

```
this.overdraftLimit = overdraftLimit
```


**Q5**

5. Vehicle Registration System

[BL3 – Apply]

10 marks

Q6

6. Bank Account Hierarchy [BL4

– Analyze]

10 marks

Q7

7. Hospital Patient Record

System [BL4 – Analyze]

10 marks

Q8

8. E-Commerce Product Catalog

[BL4 – Analyze]

10 marks

Q9

9. University Staff Hierarchy

[BL4 – Analyze]

10 marks

Q10

10. Smart Home Device

Controller [BL4 – Analyze]

10 marks

CLASS CurrentAccount EXTENDS BankAccount

ATTRIBUTES:

overdraftLimit : REAL

CONSTRUCTOR (accNo, balance, holder, overdraftLimit)

CALL BankAccount.CONSTRUCTOR(accNo, balance, holder)

this.overdraftLimit = overdraftLimit

METHOD withdraw(amount : REAL)

IF amount > 0 THEN

remaining = balance - amount

IF remaining >= (-overdraftLimit) THEN

balance = remaining

PRINT "Current withdrawal successful. New balance: ", balance

ELSE

PRINT "Current: Overdraft limit exceeded"

ENDIF

ENDIF

ENDCLASS



Save Answer